ELECTRIC LOCOMOTIVE MAINTENANCE MANUAL

Class: 3-Phase IGBT, 25 kV AC, 9000 HP Freight Locomotive (generic specification)

Document No.	ELM/MAINT/3PH-9K/Rev.04
Applicable Shed	Electric Loco Shed (fictional)
Revision Date	01 September 2026
Supersedes	Rev.03 dated 12 March 2024

1. General Technical Data

Parameter	Value
Supply voltage	25 kV AC, 50 Hz single phase
Continuous power at wheel	9000 HP (6120 kW)
Axle arrangement	Co-Co
Maximum speed	120 km/h
Starting tractive effort	46 tonnes (460 kN)
Axle load	22.5 tonnes
Gear ratio	92:17
Wheel diameter (new / condemning)	1092 mm / 1014 mm
Traction motor	6 x 3-phase induction, 1150 kW each
Regenerative braking	Yes, up to 60% of tractive effort

2. Maintenance Schedule

Schedule	Periodicity	Man-hours	Key Activities
Trip (TS)	Every trip / 5,000 km	3	Visual check of pantograph, roof equipment, bogie, brake rigging
IA	30 days +/- 3	8	Pantograph carbon strip, sanding, auxiliary compressor oil
IB	60 days +/- 5	14	Filters, cooling blower, earthing brush, brake blocks
IC	180 days +/- 10	28	Traction motor bearing grease, gear case oil, converter insulation test
Quarterly (Q)	90 days	18	Circuit breaker contact wear, transformer oil level, battery electrolyte
Annual (AOH)	12 months	76	Full roof gear overhaul, bogie examination, protection relay calibration
IOH	18 months	160	Traction motor megger, suspension, brake cylinders, wheel profiling
POH	6 years	520	Complete dismantling, traction motor rewinding check, converter overhaul

3. Pantograph Maintenance Procedure (IA Schedule)

3.1 Safety precautions. Ensure the locomotive is stabled on a track with OHE isolated and earthed. Obtain a Permit-to-Work (PTW) from the Traction Distribution (TrD) controller. Lower the pantograph and confirm using the roof inspection platform. Isolate the main circuit breaker (VCB) and apply the key interlock.

3.2 Carbon strip inspection. Measure remaining carbon thickness at five points along each strip. Replace the strip if thickness is less than 8 mm at any point, or if cracks exceed 20 mm in length, or if there is a step wear greater than 3 mm.

3.3 Contact force. Using the calibrated dynamometer, record static contact force at working height 4.5 m. Acceptable range is 70 N +/- 10 N (rising) and 60 N +/- 10 N (falling).

4. Fault Diagnosis Quick Reference

Fault Code	Description	Probable Cause	Corrective Action
F-101	VCB fails to close	Low air pressure (< 5.5 kg/cm2); interlock open	Check MR pressure, reset key, inspect interlock contacts
F-214	Converter over-temperature	Cooling blower failure; choked filter	Replace filter, check blower motor, inspect coolant level
F-307	Traction motor earth fault	Insulation breakdown; moisture ingress	Megger test (min. 5 MOhm at 1000 V), isolate bogie, replace motor
F-422	Wheel slip detected persistently	Sander blocked; sensor fault	Clear sander, check speed sensor gap (1.0 +/- 0.2 mm)
F-509	Brake pipe pressure low	Leaking BP coupling; faulty A-9 valve	Test with soap solution, replace gasket, overhaul valve

5. Torque and Clearance Values

Component	Fastener	Torque (Nm)	Remarks
Traction motor suspension	M30 bolt	1100	Apply thread locking compound
Axle box cover	M20 bolt	450	Cross-pattern tightening
Pantograph base insulator	M16 bolt	190	Check for cracked porcelain
Brake caliper mount	M24 bolt	780	Re-torque after 500 km